

Notas complementarias

MAT5508 (10182) Modelos Lineales Generalizados

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1 Métodos para el análisis de datos binarios

La varianza de los estimadores de máxima verosimilitud los obtenemos a partir de la inversa de la matriz de información. La matriz de información está dada por

$$I = I(\alpha, \beta) = \begin{bmatrix} I_{\alpha\alpha} & I_{\alpha\beta} \\ I_{\beta\alpha} & I_{\beta\beta} \end{bmatrix} = \begin{bmatrix} -\frac{\partial^2 \ell}{\partial \alpha^2} & -\frac{\partial^2 \ell}{\partial \alpha \partial \beta} \\ -\frac{\partial^2 \ell}{\partial \beta \partial \alpha} & -\frac{\partial^2 \ell}{\partial \beta^2} \end{bmatrix}.$$

En este caso, tenemos que las entradas de I son

$$\begin{aligned} I_{\alpha\alpha} &= - \left[-n_1 \frac{e^{\alpha+\beta}(1+e^{\alpha+\beta}) - e^{\alpha+\beta}e^{\alpha+\beta}}{(1+e^{\alpha+\beta})^2} - n_2 \frac{e^{\alpha}(1+e^{\alpha}) - e^{2\alpha}}{(1+e^{\alpha})^2} \right] \\ &= n_1 \frac{e^{\alpha+\beta}}{(1+e^{\alpha+\beta})^2} + n_2 \frac{e^{\alpha}}{(1+e^{\alpha})^2} \\ I_{\alpha\beta} = I_{\beta\alpha} &= - \left[-n_1 \frac{e^{\alpha+\beta}(1+e^{\alpha+\beta}) - e^{\alpha+\beta}e^{\alpha+\beta}}{(1+e^{\alpha+\beta})^2} \right] \\ &= n_1 \frac{e^{\alpha+\beta}}{(1+e^{\alpha+\beta})^2} \\ I_{\beta\beta} &= - \left[-n_1 \frac{e^{\alpha+\beta}(1+e^{\alpha+\beta}) - e^{\alpha+\beta}e^{\alpha+\beta}}{(1+e^{\alpha+\beta})^2} \right] \\ &= n_1 \frac{e^{\alpha+\beta}}{(1+e^{\alpha+\beta})^2}. \end{aligned}$$

La inversa de la matriz de información es

$$I^{-1} = \frac{1}{I_{\alpha\alpha}I_{\beta\beta} - I_{\alpha\beta}I_{\beta\alpha}} \begin{bmatrix} I_{\beta\beta} & -I_{\alpha\beta} \\ -I_{\beta\alpha} & I_{\alpha\alpha} \end{bmatrix} = \begin{bmatrix} I^{\alpha\alpha} & -I^{\alpha\beta} \\ -I^{\beta\alpha} & I^{\beta\beta} \end{bmatrix}$$

Nos interesa particularmente la entrada (β, β) de I^{-1} , la cual denotamos por $I^{\beta\beta}$, y es

$$I^{\beta\beta} = \frac{I_{\alpha\alpha}}{I_{\alpha\alpha}I_{\beta\beta} - I_{\alpha\beta}I_{\beta\alpha}} = \left(\frac{I_{\alpha\alpha}I_{\beta\beta} - I_{\alpha\beta}I_{\beta\alpha}}{I_{\alpha\alpha}} \right)^{-1} = \left(I_{\beta\beta} - I_{\alpha\beta}I_{\alpha\alpha}^{-1}I_{\beta\alpha} \right)^{-1}.$$

Demostraremos que, en este caso,

$$\text{var}(\beta) = I^{\beta\beta} = \frac{1}{\mathbb{E}(Y_1)} + \frac{1}{\mathbb{E}(n_1 - Y_1)} + \frac{1}{\mathbb{E}(Y_2)} + \frac{1}{\mathbb{E}(n_2 - Y_2)}, \quad (1.1)$$

la cual evaluada en $\hat{\alpha}, \hat{\beta}$ resulta en

$$\text{var}(\hat{\beta}) = I^{\beta\beta}(\hat{\alpha}, \hat{\beta}) = \frac{1}{y_1} + \frac{1}{n_1 - y_1} + \frac{1}{y_2} + \frac{1}{n_2 - y_2}. \quad (1.2)$$

Proof. Primero notemos que con la parametrización $\beta = \log \psi$ y $\alpha = \log\{\pi_i/(1 - \pi_i)\}$ podemos escribir

$$\begin{aligned} I_{\alpha\alpha} &= n_1 \frac{e^{\alpha+\beta}}{(1 + e^{\alpha+\beta})^2} + n_2 \frac{e^{\alpha}}{(1 + e^{\alpha})^2} \\ &= n_1 \frac{\psi\pi_2/(1 - \pi_2)}{[1 + \pi_1/(1 - \pi_1)]^2} + n_2 \frac{\pi_2/(1 - \pi_2)}{[1 + \pi_2/(1 - \pi_2)]^2} \\ &= n_1 \frac{\pi_1/(1 - \pi_1)}{[1/(1 - \pi_1)]^2} + n_2 \frac{\pi_2/(1 - \pi_2)}{[(1/(1 - \pi_2))]^2} \\ &= n_1\pi_1(1 - \pi_1) + n_2\pi_2(1 - \pi_2) \\ I_{\alpha\beta} = I_{\beta\alpha} &= n_1 \frac{e^{\alpha+\beta}}{(1 + e^{\alpha+\beta})^2} = n_1\pi_1(1 - \pi_1) \text{ y} \\ I_{\beta\beta} &= n_1 \frac{e^{\alpha+\beta}}{(1 + e^{\alpha+\beta})^2} = n_1\pi_1(1 - \pi_1). \end{aligned}$$

Entonces tenemos

$$\begin{aligned}
 [I^{-1}]_{\beta\beta} &= I^{\beta\beta} = I^{\beta\beta}(\alpha, \beta) = [I_{\beta\beta} - I_{\beta\alpha}I_{\alpha\alpha}^{-1}I_{\alpha\beta}]^{-1} \\
 &= \left[n_1\pi_1(1 - \pi_1) - \frac{(n_1\pi_1(1 - \pi_1))^2}{n_1\pi_1(1 - \pi_1) + n_2\pi_2(1 - \pi_2)} \right]^{-1} \\
 &= \left[n_1\pi_1(1 - \pi_1) \left(1 - \frac{n_1\pi_1(1 - \pi_1)}{n_1\pi_1(1 - \pi_1) + n_2\pi_2(1 - \pi_2)} \right) \right]^{-1} \\
 &= \left[\frac{n_1\pi_1(1 - \pi_1)n_2\pi_2(1 - \pi_2)}{n_1\pi_1(1 - \pi_1) + n_2\pi_2(1 - \pi_2)} \right]^{-1} \\
 &= \frac{n_1\pi_1(1 - \pi_1) + n_2\pi_2(1 - \pi_2)}{n_1\pi_1(1 - \pi_1)n_2\pi_2(1 - \pi_2)} \\
 &= \frac{1}{n_1\pi_1(1 - \pi_1)} + \frac{1}{n_2\pi_2(1 - \pi_2)} \\
 &= \frac{n_1(1 - \pi_1 + \pi_1)}{n_1n_1\pi_1(1 - \pi_1)} + \frac{n_2(1 - \pi_2 + \pi_2)}{n_2n_2\pi_2(1 - \pi_2)} \\
 &= \frac{1}{n_1\pi_1} + \frac{1}{n_1(1 - \pi_1)} + \frac{1}{n_2\pi_2} + \frac{1}{n_2(1 - \pi_2)} \\
 &= \frac{1}{\mathbb{E}(Y_1)} + \frac{1}{\mathbb{E}(n_1 - Y_1)} + \frac{1}{\mathbb{E}(Y_2)} + \frac{1}{\mathbb{E}(n_2 - Y_2)}.
 \end{aligned}$$

Ahora sabemos que la parametrización $\beta = \log \psi$ y $\alpha = \log\{\pi_i/(1 - \pi_i)\}$ son transformaciones 1 a 1 de π_1 y π_2 por lo que los estimadores $\hat{\pi}_1$ y $\hat{\pi}_2$ los podemos obtener a partir de $\hat{\alpha}$ y de $\hat{\alpha} + \hat{\beta} = y$ y $\hat{\alpha} + \hat{\beta}$ respectivamente. Esto es

$$\hat{\alpha} = \log \left(\frac{y_2}{n_2 - y_2} \right) = \log \left(\frac{y_2/n_2}{(n_2 - y_2)/n_2} \right) = \log \left(\frac{\hat{\pi}_2}{1 - \hat{\pi}_2} \right)$$

implica

$$\hat{\pi}_2 = \frac{y_2}{n_2}$$

y

$$\hat{\alpha} + \hat{\beta} = \log \left(\frac{y_2}{n_2 - y_2} \right) + \log \left(\frac{y_1/(n_1 - y_1)}{y_2/(n_2 - y_2)} \right) = \log \left(\frac{y_1}{(n_1 - y_1)} \right) = \log \left(\frac{\hat{\pi}_1}{1 - \hat{\pi}_1} \right)$$

implica

$$\hat{\pi}_1 = \frac{y_1}{n_1}$$

lo que resulta en

$$\begin{aligned}
 \hat{\pi}_2 = \frac{y_2}{n_2} &\implies y_2 = n_2\hat{\pi}_2 \quad y \\
 \hat{\pi}_1 = \frac{y_1}{n_1} &\implies y_1 = n_1\hat{\pi}_1.
 \end{aligned}$$

Por lo tanto, podemos estimar $[I^{-1}]_{\beta\beta}$ evaluando los valores esperados en $\hat{\pi}_1$ y $\hat{\pi}_2$,

$$\begin{aligned} I^{\beta\beta}(\hat{\alpha}, \hat{\beta}) &= \frac{1}{n_1 \hat{\pi}_1} + \frac{1}{n_1(1 - \hat{\pi}_1)} + \frac{1}{n_2 \hat{\pi}_2} + \frac{1}{n_2(1 - \hat{\pi}_2)} \\ &= \frac{1}{y_1} + \frac{1}{n_1 - y_1} + \frac{1}{y_2} + \frac{1}{n_2 - y_2}. \end{aligned}$$

□